MEDIUM

ADVANCED MATH

Equivalent expressions

01

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 ADVANCED MATH

Which of the following is equivalent to $\sqrt[4]{x^2+8x+16}$, where $x>0$?

A. $(x+4)^4$

B. $(x+4)^2$

C. $(x+4)$

D. $(x+4)^{\frac{1}{2}}$

Q2 GRID-IN ADVANCED MATH

$$2x^2+5x-12$$

If the given expression is rewritten in the form $(2x-3)(x+k)$, where k is a constant, what is the value of k ?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3 ADVANCED MATH

Which expression is equivalent to $\frac{h^{15}q^7}{h^5q^{21}}$, where $h > 0$ and $q > 0$?

A. $\frac{h^{10}}{q^{14}}$

B. $\frac{h^3}{q^3}$

C. $h^{10}q^{14}$

D. h^3q^3

Q4 GRID-IN ADVANCED MATH

$$\left(5x^3 - 3\right) - \left(-4x^3 + 8\right)$$

The given expression is equivalent to $bx^3 - 11$, where b is a constant. What is the value of b?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

ADVANCED MATH

If $a = 4k + 5r$ and $b = 7k - 12r + 3$, which expression is equivalent to $a - b$?

A. $-3k + 17r + 3$

B. $-3k + 17r - 3$

C. $-3k - 7r - 3$

D. $-3k - 7r + 3$

Q6 ADVANCED MATH

Which expression is equivalent to $8x^3 + 8 - x^3 - 2$?

A. $8x^3 + 6$

B. $7x^3 + 10$

C. $8x^3 + 10$

D. $7x^3 + 6$

Q7 ADVANCED MATH

$$f(x) = x^2 + bx$$

$$g(x) = 9x^2 - 27x$$

Functions f and g are given, and in function f, b is a constant. If $f(x) \cdot g(x) = 9x^4 - 26x^3 - 3x^2$, what is the value of b?

A. -26

B. $-\frac{26}{9}$

C. $\frac{1}{9}$

D. 9

Q8

ADVANCED MATH

Which expression is equivalent to $6x^8y^2 + 12x^2y^2$?

A. $6x^2y^2 2x^6$

B. $6x^2y^2x^4$

C. $6x^2y^2x^6 + 2$

D. $6x^2y^2x^4 + 2$

Q9 ADVANCED MATH

Which of the following expressions is equivalent to $8x^{10} - 8x^9 + 88x$?

A. $x7x^{10} - 7x^9 + 87x$

B. $x8^{10} - 8^9 + 88$

C. $8xx^{10} - x^9 + 11x$

D. $8xx^9 - x^8 + 11$

Q10

ADVANCED MATH

Which of the following is equivalent to $(1-p)(1+p+p^2+p^3+p^4+p^5+p^6)$?

A. $1-p^8$

B. $1-p^7$

C. $1-p^6$

D. $1-p^5$